Shiyu Chen

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Robotics perception — LiDAR–camera fusion & calibration · 3D semantic mapping · C++/ROS 2 · deep learning on edge GPUs

EDUCATION

Carnegie Mellon University — M.S. in Mechanical Engineering (Research) Aug 2025 – May 2027 (expected)
Computational Engineering & Robotics Lab (CERLAB), advisor Prof. Kenji Shimada Pittsburgh, PA

University of Cincinnati & Chongqing University — B.S. / B.Eng. in Mechanical Engineering (joint program) 2020 – 2025
Robotics Minor · GPA 3.71/4.0 (UC) · Dean's List ×4 Cincinnati, OH / Chongqing, CN

SKILLS

Languages	C++, Python, MATLAB
Robotics	ROS 1/2, Isaac Sim, Spot SDK, SLAM, LiDAR–camera calibration, stereo vision, viewpoint & path planning
Perception / ML	PyTorch, TensorRT, OpenCV, YOLO, SAM2, VLM / VLA, point-cloud processing (ICP)
Hardware / Tools	Jetson Xavier NX / AGX Orin, Ouster LiDAR, Linux, Docker, Git, SolidWorks, 3D printing

EXPERIENCE

Graduate Research Assistant, CERLAB, Carnegie Mellon University Oct 2025 – Present · Pittsburgh, PA

Semantic exploration for a legged inspection robot (CMU × YKK factory-automation project)

- Developed **5-DoF viewpoint planning** for Spot: body pitch/roll on top of planar motion reveals surfaces a level camera misses.
- Built 3D semantic mapping: per-instance detection masks back-projected into 3D object boxes in the occupancy map.
- Independently ran the full YOLOv8-seg pipeline: data collection, SAM2-assisted labeling, training and tuning, deployment.
- Sampled body postures by expected coverage gain; aim-aligned execution avoids needless reorientation.
- Added VLM-assisted viewpoint pruning (observation history + bird's-eye-view map), with a geometric fallback.
- Built an Isaac Sim testbed: three industrial scenes, ground-truth surfaces, plug-in planners, Dockerized; planner in C++/ROS.
- **+8–10 pp coverage**, **17–32% less exploration time** vs. the 3-DoF baseline; **53–73% fewer postures** than other 5-DoF methods.
- Deployed and validated on a real Boston Dynamics Spot in a machine shop.

Real-time LiDAR–camera fusion on Spot

- Calibrated the intrinsics of two cameras with unknown optical models (Spot CAM, Insta360 X4) in OpenCV.
- **Targetless registration** of Ouster OS1 point clouds and Insta360 X4 panoramas: extrinsics and transform jointly optimized.
- **10 FPS** at full LiDAR rate, **<100 ms** latency, **<10 px** reprojection error.
- Iterated a 3D-printed rigid sensor mount until registration stayed stable under gait vibration.

Shoe-interior 3D scanner (Sep 2026 – Present · sole researcher)

- Geometry-only 3D reconstruction of the inside of a shoe for fit checks (e.g., diabetic feet); scope extended to garments.
- Route A (stereo): **endoscopic probe**, two global-shutter cameras, 34 mm baseline, 148° FOV, LED ring, wall tags.
- Route B (laser): laser ring + one wide-angle camera (**laser triangulation**); both heads share one rail carriage.
- Designed the 3D-printed rail fixture and a test plan that uses the CAD model as ground truth.

Computer Vision Engineer (Intern), CISDI Information Technology May 2022 – Dec 2023 (3 co-op terms) · Chongqing, China

- Developed belt-conveyor inspection algorithms for steel plants (material type/state, particle size), mainly segmentation.
- Added GAN refinement of segmentation maps, improving material-state accuracy; compensated lens distortion.
- Proposed a Wasserstein-distance test for the particle-size algorithm, replacing manual comparison.
- Independently built a YOLOv5s object detector that supplies control signals for automatic converter tapping.
- Sped up a detector's TensorRT inference by **~40%**; ported an algorithm to C++ for DeepStream on Jetson NX / Orin.
- Built a team-wide annotation tool (own classification and video-frame modules); led 5 interns labeling ~100k images.
- Co-inventor on **4 granted computer-vision invention patents**; project featured on China Central Television.

Undergraduate Researcher, University of Cincinnati (Prof. Janet Dong) Sep 2024 – May 2025 · Cincinnati, OH

Cable-driven soft robotic arm for minimally invasive surgery

- Built a wire-actuated continuum arm (**0.27 mm/step**, **230°** bending); modeled nonlinear actuation; led hardware–control integration.

PATENTS & PUBLICATIONS

Granted invention patents (China): CN117975197B · CN117218334B · CN117218393B · CN115588114B

X. Zhan*, **S. Chen***, K. Shimada, "Pose-aware Legged Robot Semantic Exploration with Omnidirectional Perception in Confined Unknown Environments," arXiv:2609.19460, 2026. Under review at ICRA 2027. *Equal contribution.

Y. Wang, C. Liu, K. Jiang, B. Wu, **S. Chen**, J. Dong, A. Ashfaq, "Innovative Flexible Robotic Arm for Enhanced Precision in Transaortic Surgical Myectomy," *ASME J. Medical Diagnostics*, 9(2): 021001, 2026.

L. Liu, **S. Chen**, Z. Tang, "Intelligent Diagnosis of Bearing Faults Based on SDP Image Fusion," *ICICML*, Chengdu, 2023.